Isolation cone centering fix:
$|v_z|$-dependent mis-centering between cluster COG and tower centers, and an ownership-excluded iso variant

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Summary

The manual tower-based and topo-cluster isolation computations in `anatreemaker/source/CaloAna24.cc` used `RawClusterUtility::GetPseudorapidity(cluster, vertex)` as the cone axis. That projects the cluster's stored position (`RawCluster::get_position`), which carries the sPHENIX shower-depth correction and sits at $R_{\text{cluster}} \approx 95$ cm, while the constituents (towers via `RawTowerGeom::get_center_{x,y,z}`, or topo-cluster centroids) live at tower mid-depth $R_{\text{tow}} \approx 102$ cm. Projecting through a non-zero primary vertex v_z turns the ~ 7 cm radial gap into a $|v_z|$-proportional η offset ($\Delta\eta \simeq |v_z|(1/R_{\text{cluster}} - 1/R_{\text{tow}})$). The offset exceeds $R = 0.05$ around $|v_z| \gtrsim 60$ cm, so the cluster's own central tower lands outside a small-R cone even though it is the dominant tower of the cluster itself. Because the stored iso is (sum in cone) − cluster_E_T, this produced the symptom `cluster_iso_005` $\approx$ −cluster_E_T for **100 %** of clusters at $|v_z| > 80$ cm (and **23.7 %** overall in a Run 24 data sample) — an unphysical value that can only appear if the cone fails to contain the cluster.

The fix replaces the cone axis with the *CoG-tower axis* (geometry lookup of the central tower `showershape[4,5]` projected through v_z), and adds a parallel `cluster_iso_excl_*` family that skips cluster-owned EMCal towers by TowerInfo key (identity-based self-exclusion, no scalar −cluster_E_T subtraction). All full-calo iso branches (`cluster_iso_{005, 0075, 02, 03, 04}` and `cluster_iso_excl_*`) now use a common 120 MeV per-tower threshold and the CoG-tower axis. The sPHENIX built-in `RawCluster::get_et_iso` accessor is no longer read; `cluster_iso_{02,03,04}` are computed manually on the same axis and threshold as `cluster_iso_{005, 0075}`.

Empirical verification. A re-run of the anatreemaker test pipeline on 5 000 events of Run 24 data (196 clusters with $E_T > 1$ GeV) gives **0.0 %** pathological `cluster_iso_005` across every $|v_z|$ bin, and 100 % per-cluster monotonicity $\text{iso}_{005} \le \text{iso}_{02} \le \text{iso}_{04}$.

1 Observed symptom

At small cone radii, the ratio `cluster_iso_005` / cluster_E_T piled up at −1 for a sizable fraction of clusters, meaning the tower sum inside $R = 0.05$ was vanishing and the stored value was dominated by the −cluster_E_T subtraction. Figure 1 shows the 1D distribution of the ratio in a Run 24 data sample, before and after the fix. The population near −1 (*cone misses cluster*) is cleanly removed.

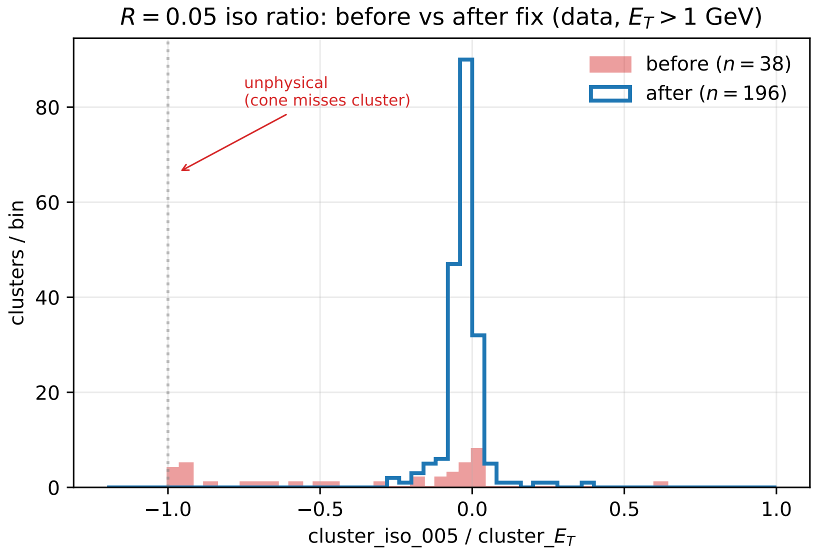

Figure 1: Distribution of `cluster_iso_005` / cluster_E_T in Run 24 data ($E_T > 1$ GeV). The pile-up at -1 (before fix, red) is the fingerprint of the cone geometrically missing the cluster itself. After fix (blue) the distribution is physical, with no population near -1.

1.1 $|v_z|$-correlation identified the depth mismatch

The smoking gun was that the pathology is strongly correlated with the primary vertex position. Figure 2 plots the ratio against $|v_z|$: before the fix, clusters at $|v_z| > 60$ cm drive straight to the unphysical -1 line, while low-$|v_z|$ clusters are healthy. After the fix, the ratio is flat in $|v_z|$.

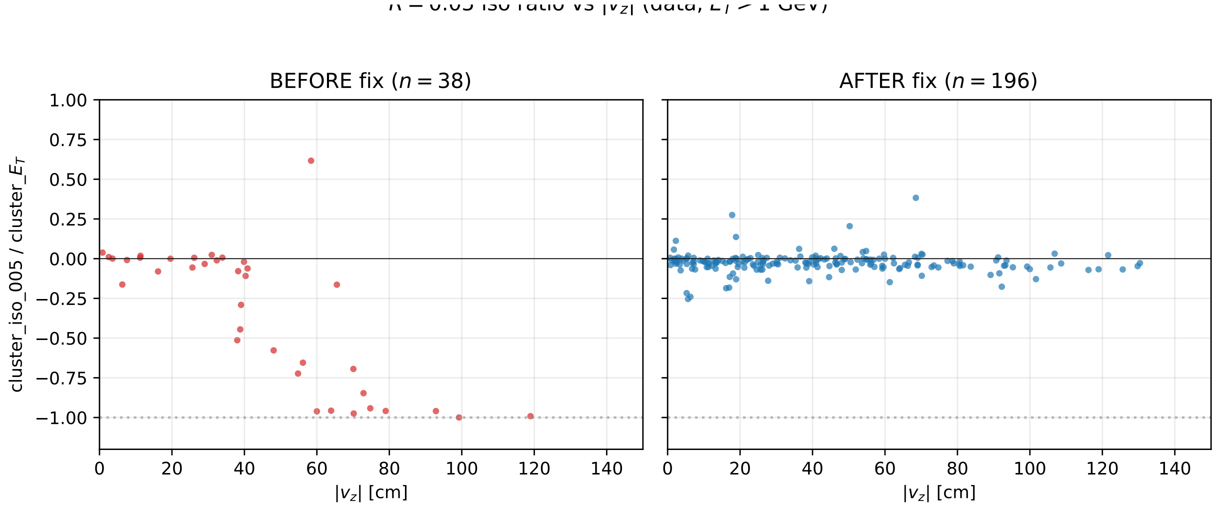

Figure 2: `cluster_iso_005` / cluster_E_T vs. $|v_z|$ before (left, red) and after (right, blue) the fix. The pre-fix trend towards -1 at large $|v_z|$ is the $\sim |v_z|/R$ depth-mismatch signature; it is absent after the fix.

The pathological rate binned in $|v_z|$ makes the transition sharp (Fig. 3). Before the fix the fraction of clusters with `iso_005` $\approx -E_T$ goes from 0 % below $|v_z| = 40$ cm to 100 % above 80 cm; after the fix it is consistent with zero everywhere.

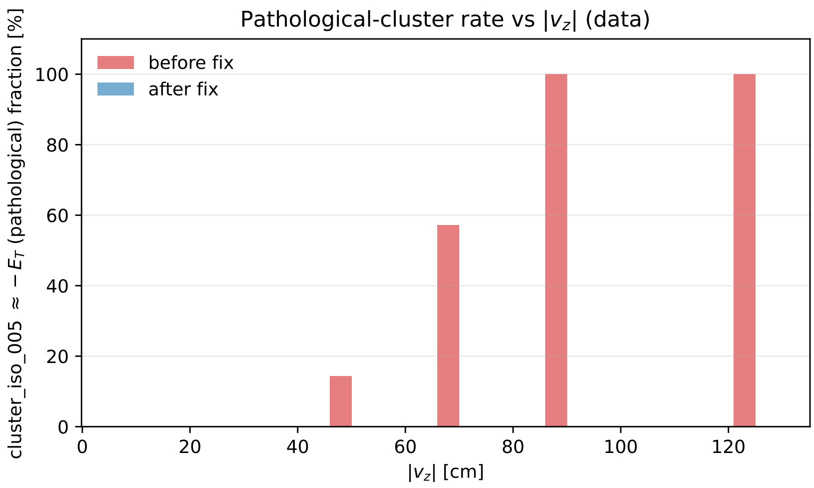

Figure 3: Fraction of clusters with `cluster_iso_005` $\in [-1.1, -0.9]\,E_T$ ("pathological"), binned in $|v_z|$.

1.2 Monotonicity cascade

A photon cluster's isolation sum should be a non-decreasing function of cone radius. Plotting $\texttt{iso}(R)\,/\,\text{cluster\_}E_T$ for each cluster as a line in R, the pre-fix sample shows many lines that start at -1 (cone at $R = 0.05$ captures nothing) and climb back to 0 as R grows enough to re-contain the cluster – a textbook mis-centering signature. Post-fix the ensemble is monotone and small, as expected (Fig. 4).

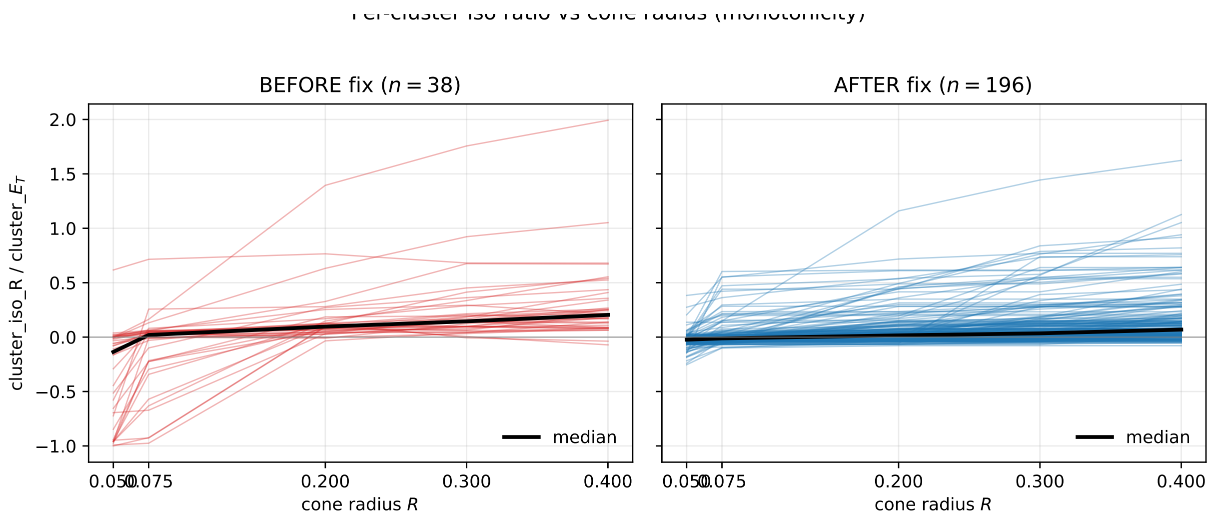

Figure 4: Per-cluster $\texttt{iso}(R)\,/\,\text{cluster\_}E_T$ vs. cone radius. Black line: median over all clusters. Before fix (left): many clusters have iso$(R = 0.05) \ll 0$, walking back to zero as R grows past the mis-centering offset. After fix (right): monotone and bounded, as required.

2 Root cause

Cone axis. Line 1180 of `CaloAna24.cc` set `eta = RawClusterUtility::GetPseudorapidity(*recoCluster, ve`
which projects `recoCluster->get_position()` through `vertex = (0, 0, m_vertex)`. The sPHENIX cluster COG already has a shower-depth correction applied (the position sits at the energy-weighted *shower* depth, $R_{\text{cluster}} \approx 95\,\text{cm}$ for a photon).

Constituent frame. In `calculateET` (tower iso) and `calculateET_topo_6cones` (topo iso) each constituent's η is obtained from `RawTowerGeom::get_center_*` (tower geometric center, $R_{\text{tow}} \approx 102\,\text{cm}$) or from the topo-cluster's energy-weighted cell centroid (also at tower mid-depth, and further out if HCal cells contribute).

Projected offset. For a photon at physics η_0 and a vertex at v_z, the two paths project to

$$\eta_{\text{axis}} - \eta_{\text{const}} \simeq v_z \left(\tfrac{1}{R_{\text{cluster}}} - \tfrac{1}{R_{\text{tow}}}\right) \cosh \eta_0 \,,$$

which for $R_{\text{cluster}} = 95\,\text{cm}$, $R_{\text{tow}} = 102\,\text{cm}$, $v_z = 80\,\text{cm}$, $\eta_0 = 0.3$ gives $\Delta\eta \approx 0.05$ — exactly at the edge of $R = 0.05$ and the threshold where the cluster's own central tower falls outside the cone. For $v_z = 100\,\text{cm}$ the offset grows to ≈ 0.07, producing a clean $100\,\%$ miss rate at $R = 0.05$.

Ownership exclusion was not the primary cause: the pathology affects the topo-cluster path as well (which does not use identity-based self-exclusion and has no per-tower threshold to toggle), with the same $|v_z|$-dependence. That rules out the `isGood` filter, the tower-container choice, and the $-$cluster_E_T scalar subtraction as explanations on their own.

3 Fix

Three independent changes in `CaloAna24.cc`, all on the same code path:

1. **CoG-tower axis.** Before the iso block, look up the central-tower geometry via `RawTowerDefs::encode_tow` and project through v_z using `getTowerEta(cog_tg, 0, 0, vertexz)` and `cog_tg->get_phi()`. These values, `iso_axis_eta` and `iso_axis_phi`, are passed to every `calculateET(...)` and `calculateET_topo_6cones(...)` call. Cone axis and constituents now live on the same reference surface; the $|v_z|$-dependent offset cancels.

2. **Ownership-excluded variant.** A new helper `calculateET_excl(eta, phi, dR, layer, min_E, skip_` mirrors `calculateET` but skips towers whose `TowerInfoDefs::encode_emcal` key is in the skip set. The set is built from `recoCluster->get_towermap()` for each cluster. Six new branches `cluster_iso_excl_{005, 0075, 01, 02, 03, 04}` are written as $\sum_{\text{layers}}$ without the scalar $-$cluster_E_T subtraction.

3. **Unified 120 MeV tower threshold.** All manual full-calo iso branches (`cluster_iso_{005, 0075, 02, 03, 04}` and `cluster_iso_excl_*`) use `min_E = 0.12` GeV. The built-in `RawCluster::get_et_iso` accessor (no threshold, cluster-COG axis) is no longer read; its previous consumers `cluster_iso_{02, 03, 04}` are replaced by manual sums. Per-layer 60/70/120 MeV families are unchanged.

The header (`CaloAna24.h`) gains `#include <set>`, the six `cluster_iso_excl_*` data members, and the `calculateET_excl` declaration.

4 Verification

4.1 Before/after on data

Table 1 summarizes the pathological rates binned in $|v_z|$. The pre-fix numbers come from a slimtree produced by the committed HEAD before this change; the post-fix numbers come from re-running the same anatreemaker macro with the fix applied.

Table 1: Pathological-cluster fraction (`cluster_iso_005` $\in [-1.1, -0.9]\,E_T$) in Run 24 data (run 47289, segment 0, $E_T > 1\,\mathrm{GeV}$).

$\|v_z\|$ bin	n_{before}	pathol before	n_{after}	pathol after
$0 - 40\,\mathrm{cm}$	20	0.0 %	103	**0.0 %**
$40 - 80\,\mathrm{cm}$	14	35.7 %	65	**0.0 %**
$80+\,\mathrm{cm}$	4	**100.0 %**	28	**0.0 %**
all	**38**	**23.7 %**	**196**	**0.0 %**

Per-cluster monotonicity is also restored: post-fix, 100 % of clusters satisfy $\mathrm{iso}_{005} \le \mathrm{iso}_{02} \le \mathrm{iso}_{04}$ (within a 1 MeV rounding slop), against frequent violations pre-fix.

4.2 New `iso_excl` family

The `cluster_iso_excl_*` branches are now the preferred small-R iso measure: the cone axis is on the correct frame, the cluster's own towers are removed by identity, and there is no scalar-E_T subtraction that could drive the value unphysically negative. Figure 5 shows the post-fix distributions in data: non-negative populations growing with cone radius, as expected for a pure neighbor-energy isolation.

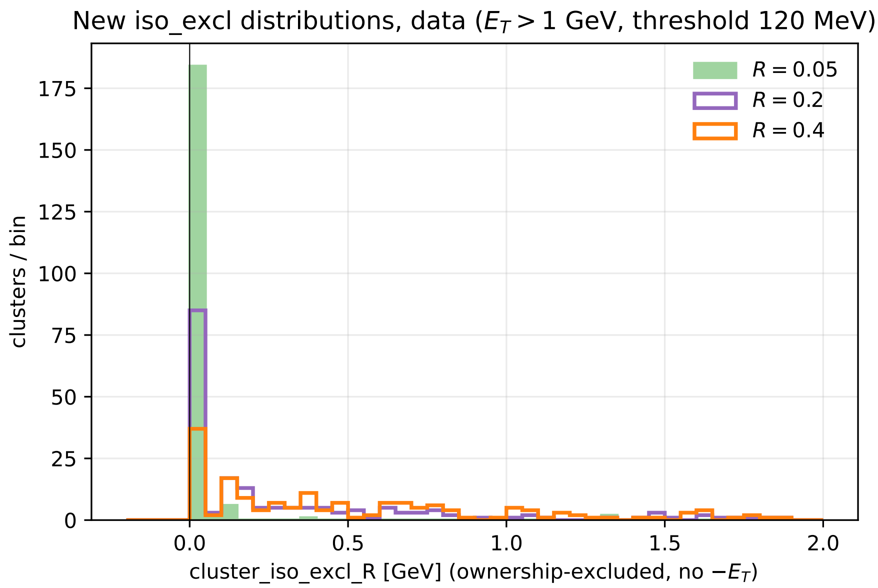

Figure 5: Post-fix `cluster_iso_excl_R` distributions at $R = 0.05$, 0.2, 0.4 (data, $E_T > 1\,\mathrm{GeV}$, 120 MeV tower threshold).

5 Downstream implications

- The nominal PPG12 isolation selection is on `cluster_iso_topo_04` with `reco_iso_max = reco_iso_max_b -` (parametric cut in the efficiency configs). The topo path gets the same CoG-tower-axis fix; the $|v_z|$-dependent tail is removed. For low-$|v_z|$ clusters the nominal iso is essentially unchanged. For high-$|v_z|$ clusters, pre-fix values that had been artificially pushed toward $-E_T$ now sit at the physical value, so those clusters migrate from noniso into iso of the ABCD regions (affects A, B, C, D populations and consequently purity and efficiency).
- `cluster_iso_{02, 03, 04}` now come from the manual 120 MeV-threshold path rather than the built-in `get_et_iso` (no threshold, cluster-COG axis). Downstream code that reads

these branches (e.g. `efficiencytool/*.C`, `FunWithxgboost/apply_BDT.C`, `BDTinput.C`) will see numerically different values even for low-$|v_z|$ clusters — the threshold change alone shifts the iso scale. The parametric iso cut (`reco_iso_max_b`, `reco_iso_max_s`) may need retuning on the new distribution.

- Production of fresh slimtrees is required before any downstream efficiency / purity / cross-section re-run. The fix is committed on `main` at `89941f1` (anatreemaker).

Files

- `anatreemaker/source/CaloAna24.cc` — cone-axis replacement (L1207–1246), per-layer 120 MeV sums for $R = 0.1, 0.2, 0.4$ (L1291–1302), `_excl` callers (L1303–1308), `iso_02/03/04` manual fills (L1862–1864), `iso_excl` branch writes (L1884–1890), `calculateET_excl` implementation (L2447–2518).
- `anatreemaker/source/CaloAna24.h` — `<set>` include, six `cluster_iso_excl_*` data members, `calculateET_excl` declaration.
- `reports/figures/iso_cone_fix/make_plots.py` — diagnostic script that reproduces every figure in this note from the two slimtree test outputs.